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(How) Does Sound Symbolism Affect Consumer's Brand Perceptions?

Submitted to: Professor of General Business Studies,
in particular Marketing (II) /
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1. Introduction

Sound symbolism, the idea that sounds and meanings have systematic relationships (Kawahara, 2020) and carry inherent meanings, has been studied extensively in linguistic and cognitive psychology. This phenomenon has also been shown to have important implications for brand perception and marketing, specifically regarding brand naming since the way consumers perceive a brand can greatly impact their purchasing decisions. A significant amount of research has been conducted exploring the use of sound symbolism in brand names to communicate brand attributes and information on a product or a service.

Although sound symbolism has received a lot of attention in the scientific research field, there is a lack of a holistic overview of research findings regarding sound symbolism in brand perception. This study provides a synopsis of 19 sound symbolism-related marketing studies published between 2000 and 2022 to address this deficit.

The overall goal of this paper is to offer an extensive literature review of sound symbolism studies in the context of brand perception and to contribute to the scientific literature in this field. Thereafter, the subgoals of this paper are to:

- provide a systematised report of sound symbolism research findings in the context of brand perception;
- identify areas not addressed by prior scientific papers to enhance further research.

Through an examination of scientific literature in this area of research, the paper also is hoped to provide a better understanding of how sound symbolism can be used to create a stronger, more effective brand, and how marketers can use this knowledge to create more effective branding and marketing materials.

Firstly, the paper presents a general overview of the research done on the topic of sound symbolism beyond the scope of marketing. Following that, the thesis focuses on the scientific literature review in the context of sound symbolism and brand perception. Finally, all themes and findings of sound symbolic marketing studies discussed will be summarised and systematised. This research is primarily based on the analysis of scientific literature with the

goal of providing a comprehensive understanding of the current state of the field and identifying areas for future research.

2. Theoretical Background & General Research in Sound Symbolism

Despite the link between sound and meaning being generally inconsistent, at times cases occur in which some sounds correlate to certain meanings. This correlation is known as sound symbolism. Over the past century, sound symbolism has been broadly studied by linguists. This chapter will provide an overview of the current state of general research in sound symbolism, including the different approaches and findings from studies in this field.

To better demonstrate the correlation between sound and meaning, a typology developed by Hinton, Nichols, and Ohala in *Introduction: Sound Symbolic Processes* (1994) will be used. According to this typology, there are four types of sound symbolism: corporeal, imitative, conventional, and synesthetic.

2.1. Corporeal sound symbolism

This type of sound symbolism represents “the use of certain sounds or intonation patterns to express the internal state of the speaker, emotional or physical” (Hinton et al. 1994, p. 2). According to the authors, corporeal sound-symbolic utterances tend to be fundamentally basic and non-segmentable vocalisations, which involve “involuntary”, and “symptomatic” sounds such as coughing or hiccupping. A study conducted by Haynie, Bower & Lapalombara in 2014 mentions high-pitched speech in a frightened state as another example of corporeal sound symbolism.

2.2. Imitative Sound Symbolism

Imitative sound symbolism, also known as “onomatopoeia”, is associated with “words or phrases representing environmental sounds” (Hinton et al. 1994, p. 3). In other words, this particular form of sound symbolism refers to the idea that certain words are formed to imitate

the sounds they represent. For example, words like “hiss”, “buzz”, and “sizzle” are thought to imitate the sounds of a snake, a bee, and frying food respectively. Imitative sound symbolism also is generally described to have a high level of linguistic structure. The article “Inanimate imitatives in English” shows how English imitative words have internal grammar (Hinton et al. 1994).

2.3. Synesthetic Sound Symbolism

This type of sound symbolism is described as “the acoustic symbolization of non-acoustic phenomena” (Hinton et al. 1994, p. 4). This way, an object’s visual, tactile, or other characteristics can be and are frequently expressed through certain phonemes or combinations of those. For example, words containing /a/ sounds are often associated with large objects and with words containing /i/ sounds with small objects (Sapir 1929). Various studies have also demonstrated the correlation between white/bright attributes and high-pitched sounds, and the linkage between black/dark attributes and low-pitched sounds respectively (Johansson /Anikin / Aseyev 2019).

2.4. Conventional Sound Symbolism

Conventional sound symbolism is “the analogical association of certain phonemes and clusters with certain meanings”, according to Hinton et al. (1994, p. 5). Conventional sound symbolism is also described as “grouping of similar meanings about similar sound” (Abelin 1999, p.15). However, this type of sound symbolism is based solely on a habitual association, compared to the previous three which are based on a natural linkage. This means specific mental perceptions are triggered by frequent pairings of phonemes, like gl- in glimmer, glisten, glitter, and glint (Elsen 2017, p. 492).

It is important to mention that in Hinton et al.’s typology, synesthetic sound symbolism represents the most crucial case in sound symbolism for the unconscious (non-arbitrary) association between sound and meaning. Synesthetic sound symbolism, compared to the other three forms, shows the linkage most vividly through characteristics like sight, taste, smell, and touch. As such, there is respectively more research on synesthetic sound symbolism, as opposed to the other types, especially in the marketing field.

Among other findings from sound symbolism-related research, sound symbolism was proved to have cross-modal correspondences with other senses, influence behaviour and development, and that it evokes specific brain responses (Lockwood / Dingemanse 2015).

3. Literature Review

This chapter aims to provide an extensive overview of the scientific literature regarding sound symbolism in the context of brand perception. Over the last two decades, various research approaches have been undertaken to study sound symbolism from different perspectives. However, due to paper limitations, this section demonstrates the results of only 19 studies in this field.

As this study aims to provide a comprehensive synopsis of the implications of sound symbolism in marketing, the literature included in the study was chosen based on the following criteria:

- only articles focusing on sound symbolism in the English language were considered;
- only marketing-related studies were used;
- since there is significantly more research on synesthetic sound symbolism, only articles examining synesthetic sound symbolism were used;
- only empirical studies including primary and/or secondary sources were examined.

This review will explore the ways in which sound symbolism imbeds can be used to influence consumer brand perceptions through examining the study findings of chosen research. In order to provide a more systematic report, the results will be grouped by specific imbeds/phonemes (e.g., consonants) to demonstrate their effects with respect to communicating brand meaning.

Additionally, the integration and combination strategies of sound symbolic brand naming will be examined. The paper will also provide a table overview of the different phonemes and their corresponding effects on brand perception (Table 1). The table will be organized by sound category (such as front/back vowels; stops/fricatives, etc.) and will include findings from the studies that observed the symbolic meaning of those sounds.

3.1. The Effects of Consonants in Brand Names on Consumers' Perceptions

While there are several ways to classify consonants, marketing-related research on sound symbolism has been observed to mainly be focused on stops/fricatives and voiced/voiceless division. Fricatives are a type of consonant that appears when air flows past the articulators (lips, teeth, tongue), creating friction (e.g., phonemes /f/, /v/, /s/ and /z/). Stops, on the other hand, are formed when the articulators are entirely closed, limiting airflow. In the English language, the letters /t/, /p/, /b/, /g/, /k/, and s are considered stops (Lowrey / Shrum, 2007, p. 42). Voiced consonants are pronounced through vibrating vocal cords (e.g., /b/, /d/ and /g/), and voiceless consonants are pronounced without vocal cord vibration (e.g., /p/, /t/ and /k/) (Lowrey / Shrum 2007, p. 42). Additionally, marketing-related sound symbolism research often aims attention at the effect the brand names containing voiced/voiceless obstruents (speech sounds produced when airflow is obstructed in the oral cavity – e.g., /b/ and /p/, /d/ and /t/, /g/ and /k/) have on the customers' perceptions (Pathak / Calvert 2020).

The usage of consonants in brand names can greatly affect customers' perception of the brand as well as the product. Research has shown that brand names containing fricative sounds, in contrast to stops, were associated with **ethicality** (Klink / Wu 2017), **lighter colours**, **angular shapes** and **smaller displayed brand marks** (Klink 2003). Klink in his study (2000) has also demonstrated the link between nonwords with fricatives and attributes like **smallness**, **sharpness**, and **softness**. The same research suggests that voiceless stops (/p/, /t/, /k/) are also related to **smallness**, **femininity**, and **lightness** in addition to **fastness** and **sharpness**, in contrast to voiced stops (Klink 2000).

The attribute of femininity, as well as masculinity, has also been examined by Guèvremont and Grohmann (2015). Their research reveals that brand **masculinity** is more effectively conveyed by incorporating stops (rather than fricatives) in the brand name. Conversely, fricatives were shown to be more effective in communicating brand **femininity** (Guèvremont / Grohmann 2015).

Another study demonstrates that any voiceless (silent) consonants in brand names (e.g., Etopal), as opposed to voiced consonants (e.g., Edopal), are perceived as more

environmentally friendly and more likely to be preferred when an environmentally friendly option is considered (Joshi / Kronrod, 2020). According to that study, brand names containing voiceless consonants (e.g., /k/, /p/, /t/) display a higher level of **modesty, purity, kindness,** and other human traits commonly associated with eco-friendliness.

A paper by Pathak, Calvert, and Lim (2020) examined the effect of voiced obstruents on perceived product /brand attributes by testing the influence of both voiced and voiceless obstruents across twenty-five hypothetical brand names in two different product categories (e.g., toilet cleaner and skin conditioner), and within the same product category (e.g., strong vs. light beer and strong vs. mild toilet cleaner). The study showed that consumers perceived brand names containing voiced obstruents as **harsh** while brand names containing voiceless obstruents were considered **soft** (Pathak et al. 2020).

A similar study by Pathak and Calvert (2020) has discovered the connection between brand names containing voiced obstruents (/b/, /d/, /g/, /z/ and /v/) and the taste of **bitterness**. In turn, voiceless obstruents (/p/, /t/, /k/, /s/ and /f/) were observed to be associated with **sweet-tasting** products (Pathak / Calvert 2020).

Additionally, medicine brand names with voiced consonants were observed to increase customers' expectations regarding the **effectiveness** of said medicine, as well as the **effect duration, price, and potential side effects**. Studies show that the presence of a voiced consonant in a brand name increased the **potency** and the **expected effectiveness** of a hypothetical medicine (Park / Motoki / Pathak / Spence 2021).

3.2. The Effects of Vowels in Brand Names on Consumers' Perceptions

Since early studies, such as Sapir's 1929 research, vowels have gotten far more attention in sound symbolism research than consonants. In the context of marketing-related research on sound symbolism, vowels are typically classified according to a front/back division. This division relates to the highest point attained by the tongue during articulation in the mouth (Maglio / Rabaglia / Feder / Krehm / Trope 2014, p. 2). As such, front vowels are developed when the tongue's highest position is near the front of the mouth (e.g., /i:/, /e/, /ɪ/ and /æ/), and

back vowels are produced when the tongue's highest point is at the back of the mouth accordingly (e.g., /ʌ/, /ɜ:/ and /ə/) (Preziosi / Coane 2017, p. 2).

While not directly related to marketing, a study done by Sapir (1929) must be mentioned since his work is one of the earliest studies done on the front/back vowel distinction effect on human perception in the English language. Moreover, his research was used as a base for a number of marketing studies to follow and, therefore, has played a significant role in exploring the effects of sound symbolism in the marketing context. In his research, Sapir used 60 pairs of fictional non-words in the consonant-vowel-consonant form (e.g., mil/mal) as “foreign” terms for words like “table” and told the research participants to decide if those terms were referring to a small or a large item. The results of the research show that the majority paired associated **large** objects with words containing front vowels and **small** objects with words containing back vowels (Sapir 1929).

The influence of front/back vowels in brand names on size perception has also been studied by Klink (2000). The research demonstrates that brand names containing front vowels were identified as being **smaller** compared to those containing back vowels, which, in turn, were perceived as **large**. In the same and later research, Klink also connects front vowels with attributes like **lightness** (opposed to both heaviness & darkness), **softness**, **femininity**, **friendliness**, **mildness**, and angular-shaped, smaller displayed brand marks (Klink 2000, 2003). Those attributes did not appear to be restricted by a specific product category (Klink 2000). The same conclusion has been reached regarding children's brand perception (Baxter / Illicic / Kulczynski / Lowrey 2015).

A study by Lowrey and Shrum (2007) has also examined the connection between vowels and size awareness. In their research, study subjects were requested to select a brand for a two-seat convertible or an SUV. As a result, participants tended to match brand names containing front vowels with **small size** and **quickness** (convertible), and brand names containing back vowels with **large size** and **slowness** (SUV), which illustrated the front & back vowel sounds' implied attributes (Lowrey / Shrum 2007).

In the same research, another study has been done, this time varying product attributes instead of the product category (cold, clean, and crisp-tasting beer compared to smooth, mellow, and rich-tasting beer). Results show that brand names containing front vowels were preferred to

describe the product with characteristics like **cold**, **clean**, and **crisp**, whereas when the attribute was **smooth**, **mellow**, and **rich**, back vowel sound words were most often chosen (Lowrey / Shrum 2007). Other research also reveals that back vowels in brand names are generally associated with **smoothness**, **creaminess**, and **richness** (Yorkston / Menon 2004), as well as **thickness** and **chewiness** (Baxter / Ilicic / Kulczynski / Lowrey 2017).

Regarding the taste factor, there has been additional research examining the connection between long/short vowel sounds and the taste of sweetness. As such, brand names containing long vowels were demonstrated to be associated with **sweetness** more than those including short vowel sounds (Pathak / Calvert 2021).

A connection between brand personality and front/back vowel presence in a brand name has been observed in a study by Klink and Athaide (2012). According to the findings of this study, adopting back vowels in brand names helps develop a **Ruggedness** brand personality. This study also discovered that brand names with front vowels convey more **Sophistication** and **Sincerity**. Back vowels, in contrast, were demonstrated to express **Down-to-earth** characteristics (Klink / Athaide 2012).

Gender preferences have also been researched within the context of marketing-related sound symbolism, specifically regarding the usage of vowel sounds in brand names. For example, **females** were observed to be more responsive and favourably inclined towards brand names containing front vowels. In contrast, **males** tended to respond more positively to brand names with back vowels (Klink 2009).

3.3. The Effects of Sound Frequency in Brand Names on Consumers' Perceptions

While there is comparatively less research on sound frequency in the context of marketing-related sound symbolism, there is growing evidence that the use of high/low -frequency sound plays a crucial role in shaping customers' perceptions of a brand. High-frequency sounds (e.g., /f/, /s/, /i/) are defined as sounds with a short wavelength in physical terms (Tsur 2006). Respectively, low-frequency sounds (e.g., /b/, /g/, /o/) have longer wavelengths.

In particular, the effects of high-frequency sounds on customers' brand perceptions have been the subject of several studies. For example, brand names containing higher frequency vowel and consonant sounds were most often associated with **smallness**, **lightness** (opposed to both heaviness & darkness), **weakness**, **softness**, **coldness**, and **femininity** (Klink 2000).

Another study has demonstrated that varying sound frequency in brand names can influence the perception of healthy food. Specifically, the research found that brand names containing higher-frequency sounds were perceived as being **healthier** than those names containing lower-frequency sounds (Motoki / Park / Pathak / Spence 2020). However, the researchers observed that the higher/lower frequency sounds created a sense of healthfulness in savoury foods but not in sweet ones.

A more recent study discovered the connection between the sounds' frequency in brand names and the evaluation & potency dimensions (Motoki / Pathak / Park / Spence 2022). The researchers used a framework of concept meanings known as the “semantic differential technique”, classifying sound symbolism into three dimensions - evaluation, potency, and activity (Motoki et al. 2022). In the study, evaluation was assessed by factors like positive-negative and good-bad; potency was measured by items such as strong-weak and heavy- light and activity was measured by items such as active-passive and fast-slow. According to Motoki et al. (2020), brand names containing high-frequency sounds were most often linked to **higher evaluation** and **lower potency**, whereas lower-frequency sounds were shown to be associated with **lower evaluation** and **higher potency**. The authors also mention the lack of substantial linkage between the activity dimension and high/low- frequency sounds (Motoki et al. 2022).

Additionally, high-frequency sounds were found to better communicate **ethicality** compared to low-frequency sounds, according to the research done by Klink and Wu (2017). Moreover, the same research has shown that brand names with high-frequency sounds might be especially beneficial for **neutralising** the impact of negative ethical information. As an example, the study compared Vittel and Contrex, bottled water brands marketed by Nestlé. In case Nestlé would be criticised for its dramatic packaging footprint, the authors claim that Vittel may be perceived as the more ethical out of the two due to the higher frequency sounds in the brand name (Klink / Wu 2017).

3.4. Incorporating Sound Symbolism: Positioning and Combination Strategies

The role of positioning the right phoneme and the combination of different phonemes can be crucial in determining the success of a brand name. Despite its potential importance, there has been limited research on the direct integration and usage of sound symbolism in brand naming. This subchapter aims to explore the practical impact of the placement, art, and combination of sound symbolism in a brand name, providing insights into the effective use of sound symbolism in brand naming by using the available literature.

Although both the vowels and consonants of brand names were found to be effective in communicating product-related information (Klink 2000), research has shown that in a brand name, vowels carry more meaning, compared to consonants (Klink / Wu 2013). Nonetheless, the same study demonstrates that integrating consistent vowel and consonant imbed combinations in a brand name has an additive effect on conveying the brand meaning and a positive influence on a product choice.

As for phoneme positioning, a study by Klink / Wu (2013) has found that inserting imbeds after the first syllable of a brand name assists in communicating the brand meaning more effectively. The research has also found that integrating multiple imbeds in a brand name is more successful in expressing brand meaning than using a single imbed (Klink / Wu 2013). Earlier research has shown that brand names using sound symbolic imbeds are generally perceived as more preferable compared to brand names not containing any sound symbols (Klink 2000, McCune 2011).

Table 1. **Key Topics and Findings of Marketing-related Sound Symbolism Research.**

Theme	Findings/Respective Attributes	Author(s)
Consonants	Smallness, sharpness, softness (fricatives) Smallness, femininity, lightness, fastness, sharpness (voiceless stops)	Klink (2000)
	Lighter colours, angular shapes, and smaller displayed brand marks (fricatives)	Klink (2003)
	Masculinity (stops), femininity (fricatives)	Guèvremont / Grohmann (2015)
	Ethicality (fricatives)	Klink / Wu (2017)
	Eco-friendliness, modesty, purity, kindness (silent consonants)	Joshi / Kronrod (2020)
	Harshness/softness (voiced/voiceless consonants respectively)	Pathak et al. (2020)
	Bitterness/Sweetness (voiced/voiceless obstruents respectively)	Pathak / Calvert (2020)
	Increased the potency and the expected effectiveness (voiced consonants)	Park et al. (2021)
Vowels	Smallness, lightness, softness, femininity, friendliness, mildness, and angular-shaped, smaller displayed brand marks (front vowels). Largeness (back vowels)	Klink (2000, 2003)
	Smoothness, creaminess, richness (back vowels)	Yorkston / Menon. (2004), Lowrey / Shrum (2007)
	Smallness, quickness, cold, cleanliness, crispiness (front vowels). Largeness (back vowels)	Lowrey / Shrum (2007)

	Female/male preferability (front/back vowels respectively) Sophistication and Sincerity/Ruggedness and Down-to-Earth (front/back vowels respectively) Smallness/largeness in children's perspective (front/back vowels respectively) Thickness, chewiness (back vowels) Sweetness (long vowels)	Klink (2009) Klink / Athaide (2012) Baxter et al. (2015) Baxter et al. (2017) Pathak / Calvert (2021)
Frequency	Smallness, lightness, weakness, softness, coldness, and femininity (high frequency) Ethicality (high frequency) Healthiness in savoury foods High/low evaluation, low/high potency (high/low frequency respectively)	Klink (2000) Klink / Wu (2017) Motoki et al. (2020) Motoki et al. (2022)
Incorporating Sound Symbolism	Both vowels and consonants are effective in communicating product-related information Sound symbolic brand names are seen as more favourable than those without imbeds Vowels carry more inherent meaning. Embedding after first syllable improves brand meaning communication. Integrating multiple imbeds in a brand name is more successful in expressing brand meaning	Klink (2000) McCune (2011), Klink (2001) Klink / Wu (2013)

4. Conclusion

The purpose of this paper was to compile the existing evidence on the effects of sound symbolism in brand naming and customer perception. To achieve that objective, a number of marketing—related articles on sound symbolism were examined and analysed to determine the current state of knowledge on the topic. Following this, a systematic literature review was created, categorised by type of phonemes (consonants, vowels, etc.) to determine the effects of specific imbeds on customers' brand perception.

In order to create a comprehensive report, only empirical studies based on primary and/or secondary sources with an emphasis on synesthetic sound symbolism in the English language were considered. Additionally, any studies with no relation to marketing were excluded from the review. The selected publications were then examined in terms of methodology and findings.

The results of the studies reviewed indicate that sound symbolism does have an influence on customers' brand perceptions. The findings suggest that products have a higher degree of attractiveness if their brand name includes sound-symbolic phonemes, and, therefore, are generally preferred over those without any imbeds (Klink, 2001; McCune, 2011).

Additionally, certain phonemes have been shown to carry inherent meanings and strong associations in consumers' minds. For instance, the reviewed literature has demonstrated that fricatives are associated with ethicality (Klink & Wu, 2017), and back vowels are associated with attributes like smoothness, creaminess, and richness (Yorkston et al., 2004; Lowrey & Shrum, 2007). In other words, studies have discovered a linkage between particular phonemes and several specific attributes.

The brand names using sound symbolism were thus shown to transmit meaning different from their literal definition, and this inherent meaning was proved to be able to affect consumer perceptions and decision-making. This shows the significance of sound symbolism's implications for branding strategies. Understanding the impacts of sound symbolism and using it in brand naming strategies may assist marketers in developing stronger and more effective brands that stand out in a competitive marketplace.

This paper should be considered in light of its limitations. First, all of the studies reviewed were conducted in English, and it is unclear whether the results would generalize to other linguistic and cultural contexts. Additionally, this study did not consider the distinction between native and non-native English-speaking customers, as well as their proficiency level. Hence, the possible difference in effects is unsettled. Second, further research is needed to determine the potential effects other forms of sound symbolism (e.g., conventional, or imitative) may have on consumer perceptions in the marketing context. Furthermore, this paper does not include any research on the mechanisms of sound symbolism due to the lack of relevant literature on this topic. The topic of sound symbolic mechanisms and how sound symbolism is processed by the brain is, therefore, out of the scope of this paper. Finally, as most studies focused on front/back vowels and voiced/silent, fricative/stop consonants distinction, existing research on sound symbolic brand naming might be expanded to examine further divisions and their inherent meanings (e.g., vowels depending on the lip rounding, consonants depending on the place of articulation).

In conclusion, the existing evidence demonstrates that sound symbolism does have a role in customer perception of brand names, as various sounds or phonemes may stimulate specific emotions and associations in people, resulting in good or negative brand images and influencing brand identification and recall. This study provides an extensive overview of the literature in this field and highlights the importance of considering sound symbolism in brand naming strategies as it can have a significant impact on customer perception and ultimately on brand success.

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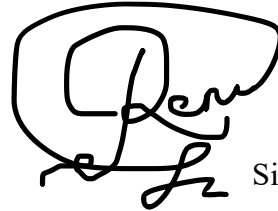
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Sworn Declaration

Declaration

I hereby affirm that I have completed the present Master's / Bachelor's thesis with the topic:
„ (How) Does Sound Symbolism Affect Consumer's Brand Perceptions? „
and did not use any aids other than those indicated. The passages which have been taken from other works, either in their wording or in their sense, have been marked as borrowings in each individual case by stating the source, including the secondary literature used.

Frankfurt (Oder), 12.02.2023

A handwritten signature in black ink, consisting of a large, stylized capital 'Q' followed by a cursive 'u' and a horizontal line with a small flourish at the end.

Signature